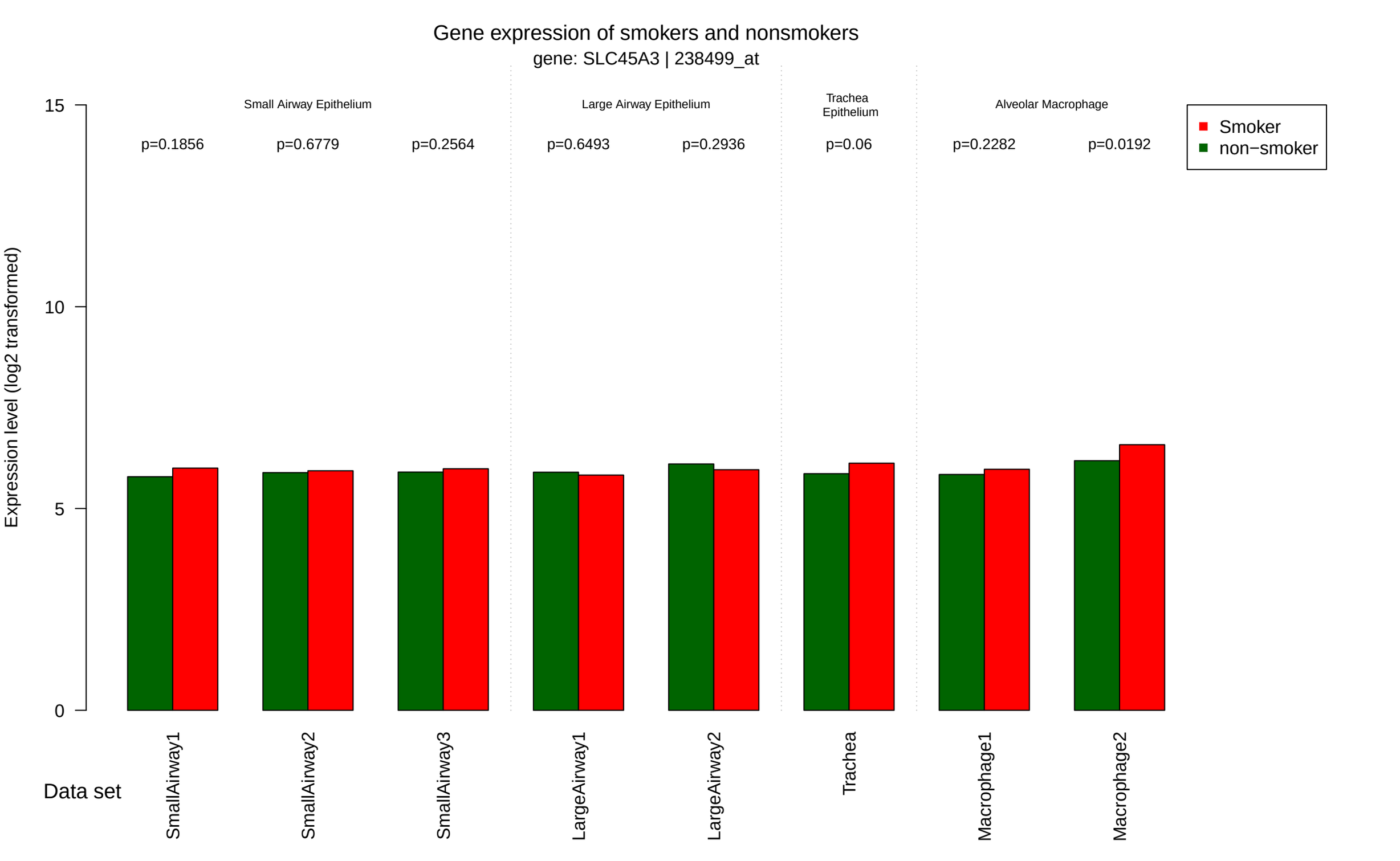
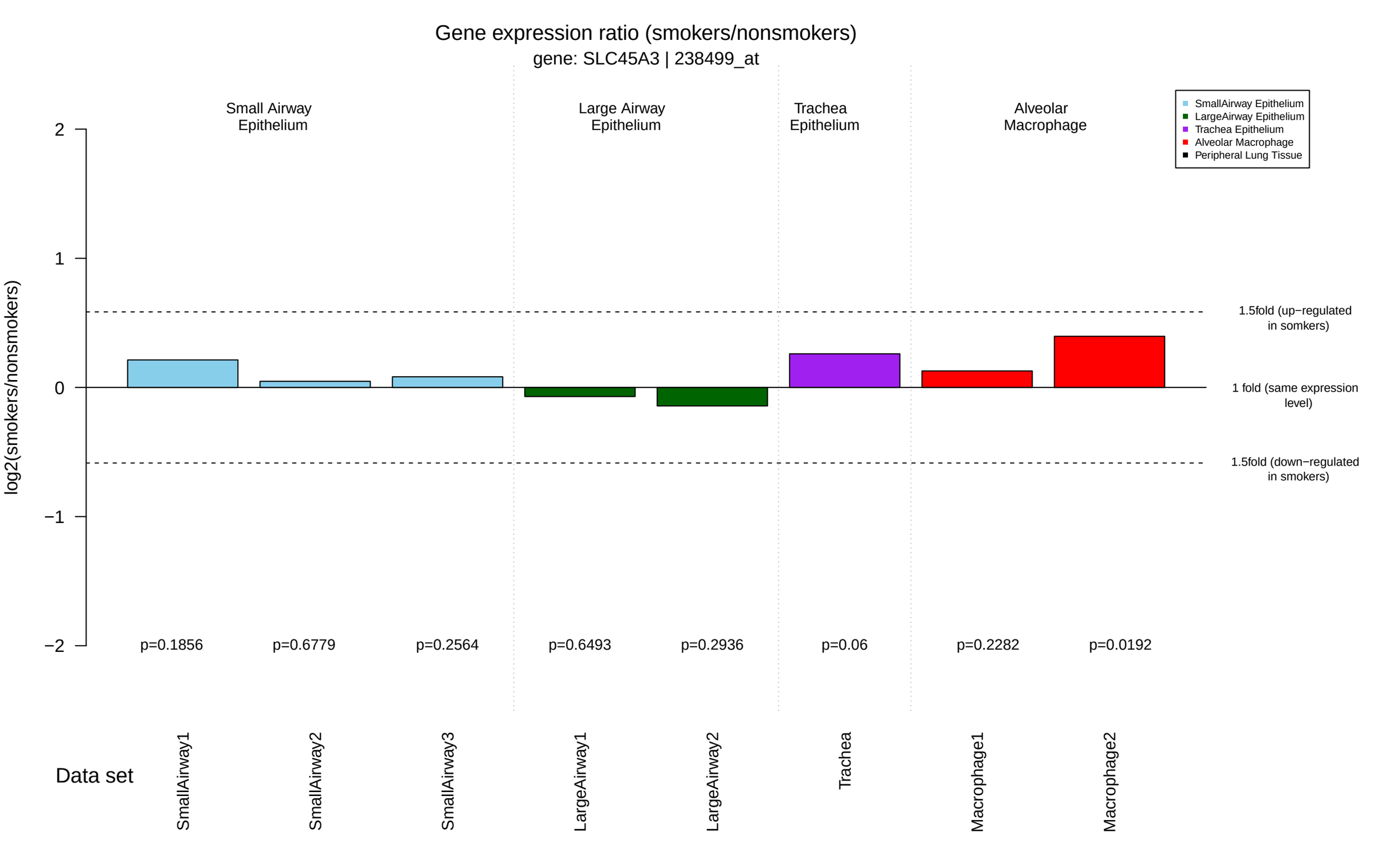
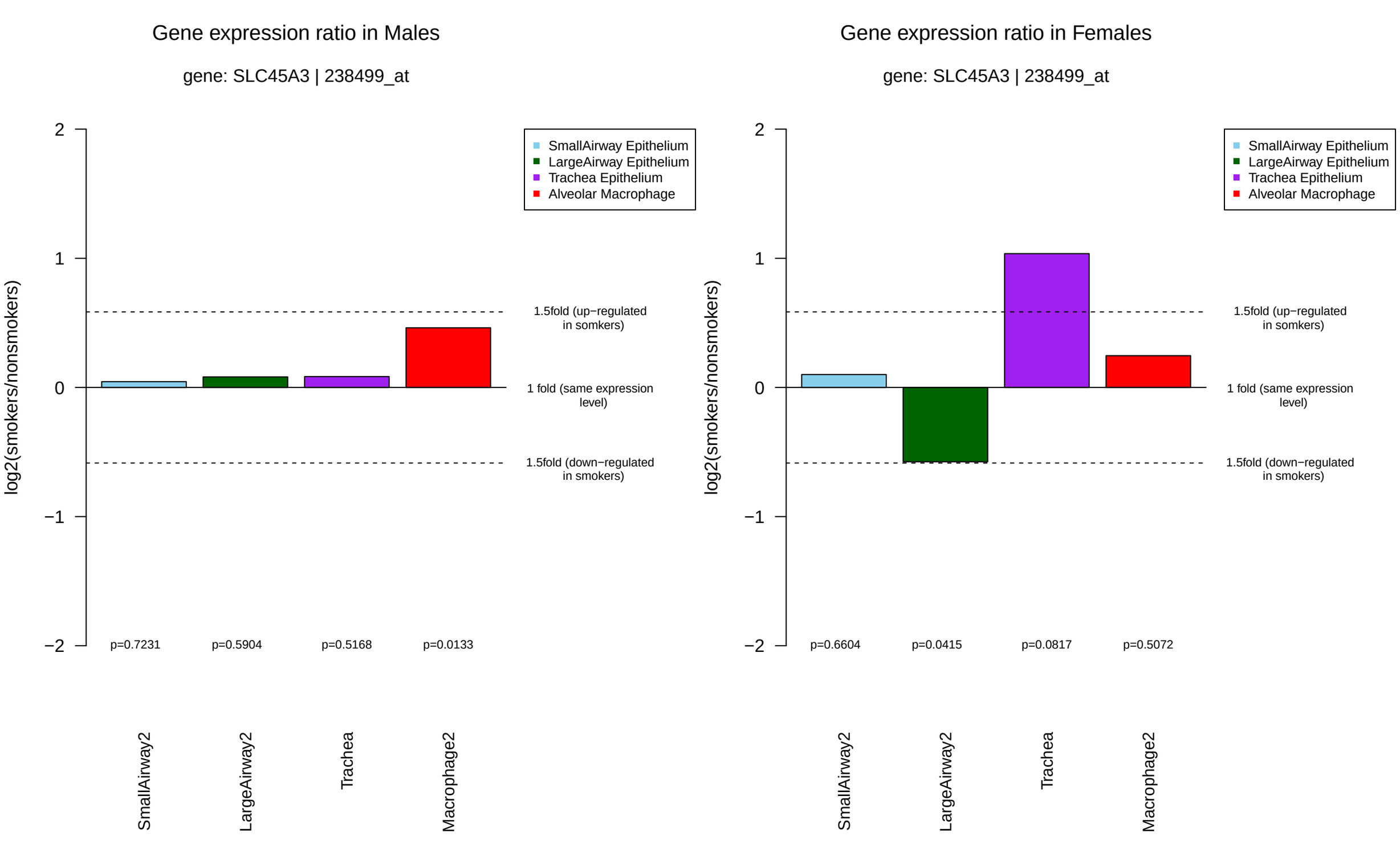
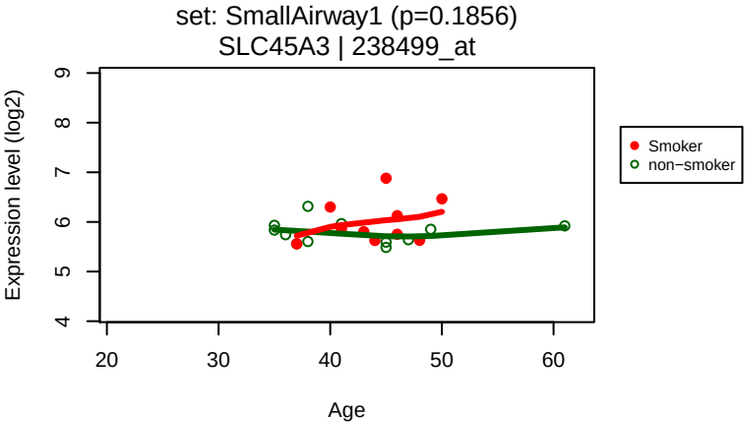
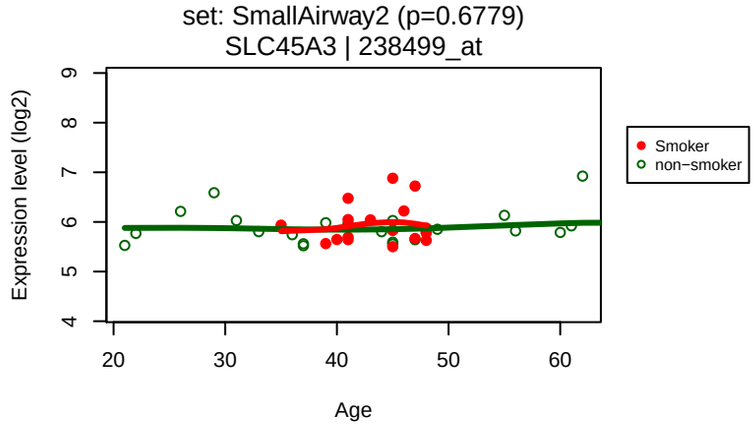
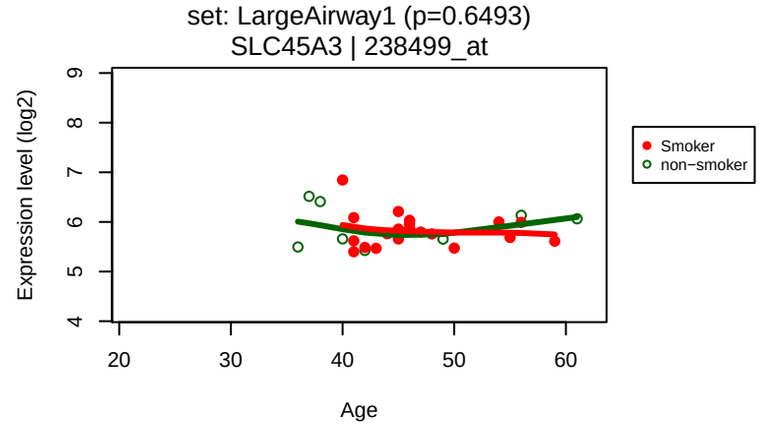

Gene expression of smokers and nonsmokers
gene: SLC45A3 | 238499_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1856
p=0.6779
p=0.2564
p=0.6493
p=0.2936
p=0.06
p=0.2282
p=0.0192
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

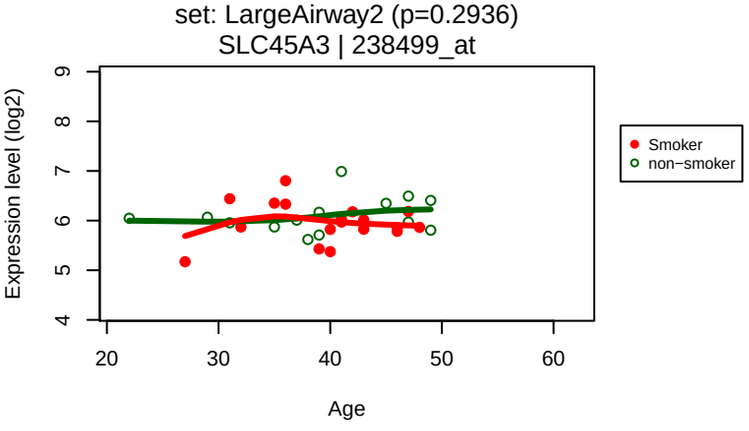

Gene expression ratio (smokers/nonsmokers)
gene: SLC45A3 | 238499_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1856
p=0.6779
p=0.2564
p=0.6493
p=0.2936
p=0.06
p=0.2282
p=0.0192
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

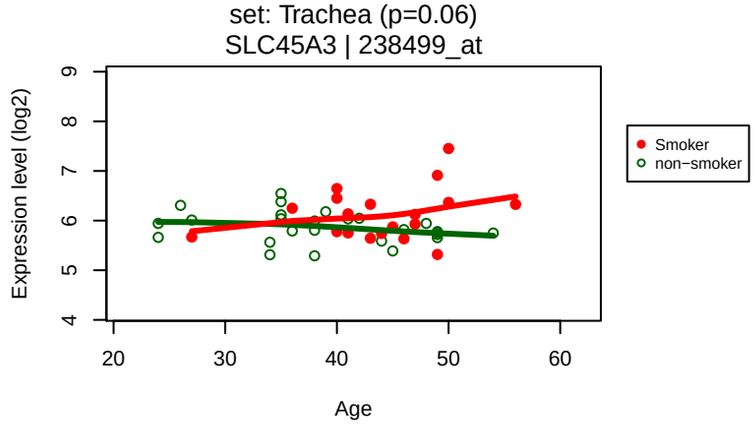

Gene expression ratio in Males
gene: SLC45A3 | 238499_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7231
p=0.5904
p=0.5168
p=0.0133
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SLC45A3 | 238499_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6604
p=0.0415
p=0.0817
p=0.5072
SmallAirway2
LargeAirway2
Trachea
Macrophage2

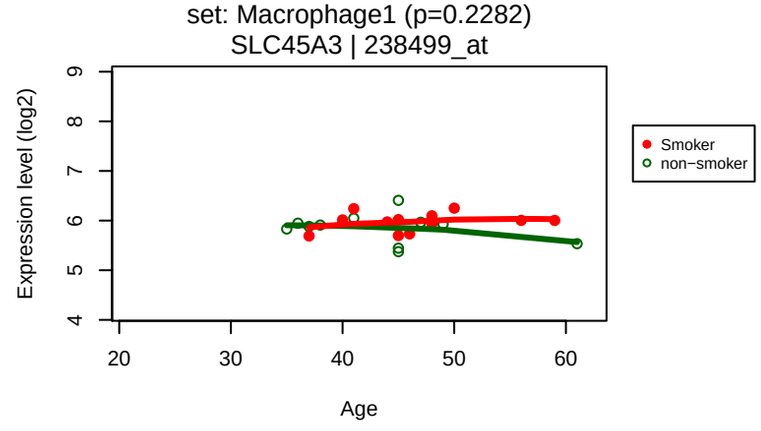

set: SmallAirway1 (p=0.1856)
SLC45A3 | 238499_at
Expression level (log2)
Age
Smoker
non−smoker

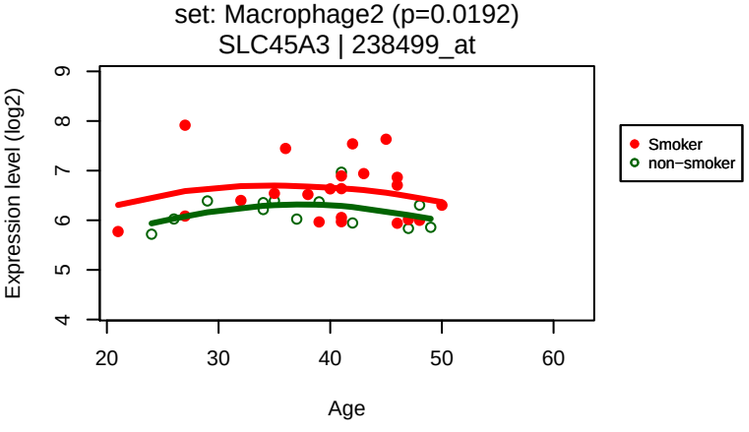

set: SmallAirway2 (p=0.6779)
SLC45A3 | 238499_at
Expression level (log2)
Age
Smoker
non−smoker

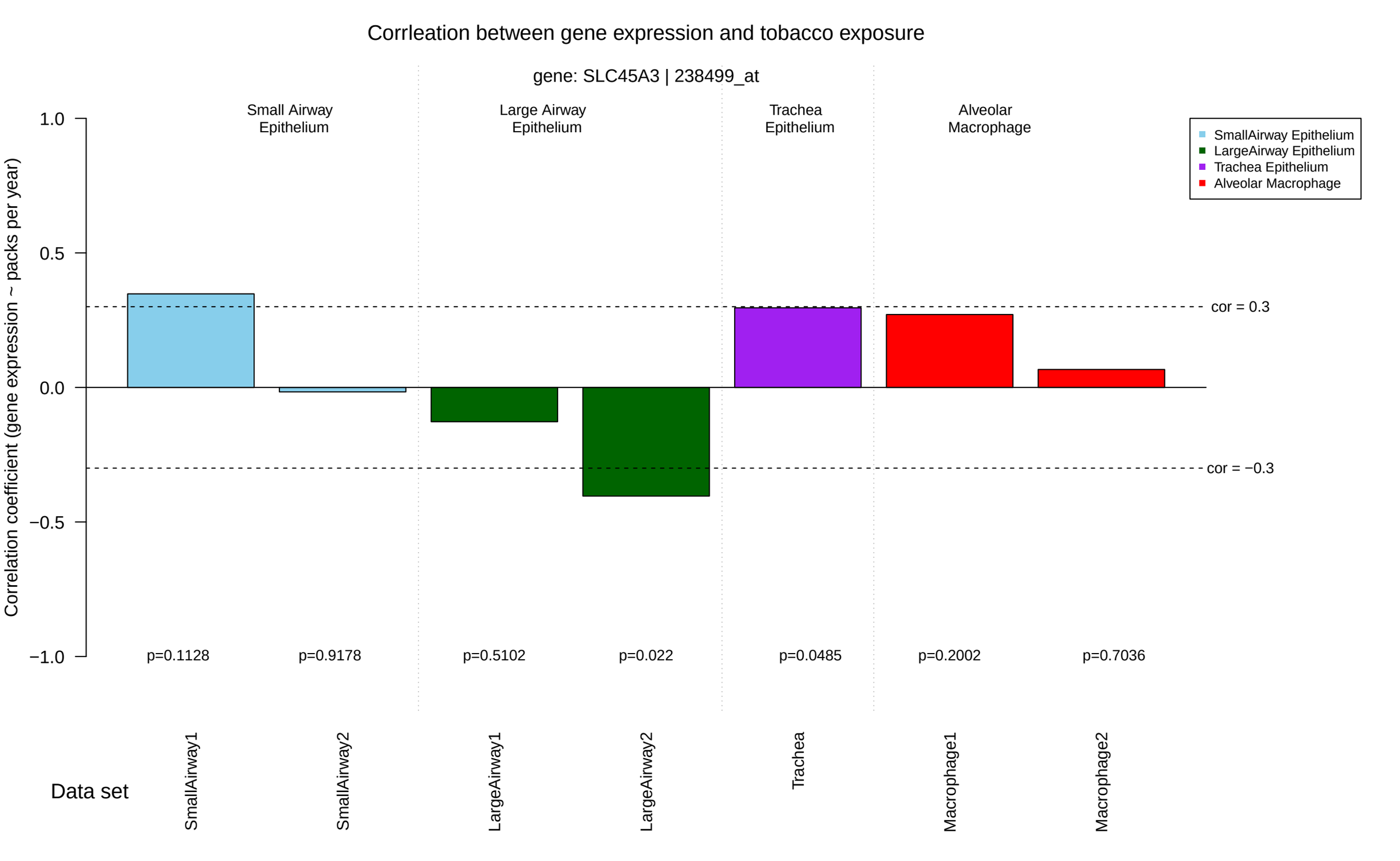

set: LargeAirway1 (p=0.6493)
SLC45A3 | 238499_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2936)
SLC45A3 | 238499_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.06)
SLC45A3 | 238499_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2282)
SLC45A3 | 238499_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0192)
SLC45A3 | 238499_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SLC45A3 | 238499_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1128
p=0.9178
p=0.5102
p=0.022
p=0.0485
p=0.2002
p=0.7036
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2